

CHLOÉ SERRE-COMBE

PhD Student in Statistics · Extreme Value Theory, Geostatistics

IMAG, University of Montpellier, France

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EDUCATION

PhD in Statistics (ongoing)

University of Montpellier, France

2022 - 2026

Master in Statistics and Data Science, specialization in Biostatistics

University of Montpellier, France

2020 - 2022

EXPERIENCE

Temporary Lecturer and Research Assistant (ATER) / PhD Student

IMAG, University of Montpellier and Inria LEMON team

2025 - 2026

Montpellier, France

PhD Candidate with Teaching Duties

IMAG, University of Montpellier and Inria, LEMON team

2022 - 2026

Montpellier, France

- Thesis topic: Stochastic generator for extreme precipitation in urban areas at high spatio-temporal resolution.
- Methods: Spatio-temporal extreme-value modeling, extreme dependence and statistical downscaling.
- Supervisors: Nicolas MEYER, Thomas OPITZ and Gwladys TOULEMONDE.

Data Science Apprentice

KanopyMed

September 2021 - August 2022

Clapiers, France

- Master thesis: Quantification and modeling of mortality related to air pollution
- Methods: Spatial statistics, epidemiology
- Supervisors: Ulysse RODTS, Grégoire MERCIER

Intern

IMAG, University of Montpellier

July 2020 - August 2021

Montpellier, France

- Enhanced the RKeOps package, a software for 2D matrix reduction computations on GPUs
- Supervisors: Benjamin CHARLIER, Ghislain DURIF

Intern

Aix-Marseille University

June 2018 - July 2018

Aix-en-Provence, France

- Miller-Rabin primality test, understanding, oral presentation and report writing
- Supervisor: Julien KELLER

PUBLICATIONS

PREPRINT

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Submitted, 2026. Preprint: arXiv:2602.19774.

IN PROCEEDINGS OF CONFERENCES

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Journées de Statistique (JDS), 2025.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. EVAN, 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. Journées de Statistique (JDS), 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modélisation statistique de précipitations urbaines à fine échelle spatio-temporelle*. Journées de Statistique (JDS), 2023.

ORAL PRESENTATIONS

INVITED SEMINARS

- May 2026** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
KIT, Karlsruhe, Germany
- Feb. 2026** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Laboratoire J.A. Dieudonné, Nice, France
- Jan. 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
PhD Seminar, IMAG, University of Montpellier, France

INTERNATIONAL CONFERENCES

- Dec. 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
SWGEM, Grenoble, France
- July 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVAN, Istituto Veneto di Scienze, Lettere ed Arti, Venice, Italy
- June 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
IMSC, Météo France, Toulouse, France
- June 2023** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVA, Bocconi University, Milan, Italy

NATIONAL

- June 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Journées des Statistiques (JDS), Aix-Marseille University, Marseille, France
- Dec. 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Occimath Seminar, IMT, Toulouse, France
- May 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Journées des Statistiques (JDS), University of Bordeaux, France
- April 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Rencontres des Jeunes Statisticiens (RJS), Porquerolles, France
- Sept. 2023** *Modeling urban precipitation at high spatio-temporal resolution*
MUFFINS Workshop, Aix-en-Provence, France
- July 2023** *Modeling urban precipitation at high spatio-temporal resolution*
Journées des Statistiques (JDS), University of Brussels, Belgium
- June 2023** *Towards statistical modeling of urban rainfall on a fine spatio-temporal scale*
RESSTE Day, FRUMAM, Marseille, France

SKILLS

Programming Languages: R, Python, HTML/CSS, JavaScript, Bash

Development Tools: L^AT_EX, Git, VS Code

Operating Systems: Ubuntu, Windows

Languages: **French** (native), **English** (fluent), **Italian** (intermediate)

TEACHING

University of Montpellier / Polytech Montpellier

Undergraduate programmes (Mathematics, Biology, Engineering years 1–4)

2022–2026

356 hours

- Statistics, Probability, Analysis and Linear Algebra
- Tutorials and practical sessions using R and Python
- Lectures, exam preparation and grading

INVOLVEMENT

Co-organizer of a working group on extreme value statistics

2024–2026

IMAG, University of Montpellier, France

Organization of a working group among PhD students, meeting every two to three months to discuss research articles in extreme value statistics.

Co-organizer of the Mathematics PhD seminar

2023–2024

IMAG, University of Montpellier, France

Organization of a biweekly doctoral seminar, including inviting speakers and managing the budget. Organization of a PhD research day with poster sessions.

Outreach activity for undergraduate women in mathematics (*Maths C pour L*)

Feb 2024

CIRM, Marseille, France

Presentation of my academic path and research activities, followed by discussions with students to promote careers in science and mathematics.